



## BEFORE

Developments in new agricultural techniques—such as the introduction of farm machinery, pesticides, and fertilizers—revolutionized food production in the 19th and early 20th centuries.

## THE ADVENT OF MECHANIZATION

By the middle of the 19th century, steam engines were driving **farm machinery**. At the end of the



TRACTOR STATION, 1930

19th century the development of oil-powered **traction engines** led to a revolution in agricultural methods, allowing greater areas to be

farmed by fewer workers. In the Soviet Union, tractor stations rented machines to nearby **collectivized** farms—small peasant holdings that were joined together into vast cooperative farms under Stalin. The policy transformed agricultural production in the USSR **<<280–81**.

## FERTILIZERS AND PESTICIDES

Decline of the natural fertility of the soil was already a major concern in the mid-19th century **<<290–91**. Both Europe and America had begun to depend on imports of **guano** (dried excrement of fish-eating birds) from South America. Guano improves crop yields by adding nitrogen and phosphorus to soil. The use of **pesticides** also increased during this period, with the discovery of pyrethrum oil and derris, both natural insecticides extracted from plants. Compounds of heavy metals, including copper and arsenic, were used to combat plant diseases.

# Feeding the World

**In the 1960s a “Green Revolution” in biotechnology boosted agricultural productivity. The efforts of agriculturalists and scientists resulted in improved fertilization and pest controls and the development of specially bred strains of high-yield, disease-resistant crops.**

**T**he founding of the United Nations’ Food and Agriculture Organization (FAO) in 1945 signaled an international desire to create change in the postwar era through a sharing of knowledge and resources to improve crop performance and eliminate hunger. The FAO focused special attention on rural areas in developing nations, as this is where the majority of the poor and hungry were, and still are, located.

## New methods

In the 1940s the biochemist Norman Borlaug (see right) initiated an experimental program to assist

poor Mexican farmers in increasing their wheat production. Borlaug concentrated on plant breeding. Over a 20-year period he developed a strain of high-yield dwarf wheat that was resistant to a variety of diseases and pests. Borlaug’s wheat was cultivated in India and Pakistan with spectacular results. In 1960 the International Rice Research Institute was established in the Philippines to improve the production of rice, the staple diet of much of the world’s population. The institute’s work has helped rice farmers increase production by an average of 2.5 percent each year since 1965. These transformations in agriculture became known as the Green Revolution. Research continues to develop a range of rice varieties or “cultivars” to suit the needs of different regions including

**INDIA AND PAKISTAN almost doubled their wheat production between 1965 and 1970.**

New Rice for Africa (NERICA)—a strain of rice suited to Africa’s dry ecosystems

In the 1960s the insecticide DDT was the first of a range of new organic chemical insecticides that were widely used for their capacity to control multiple species of pests with a single treatment. These chemicals were put to

use in developing countries. However, in 1962 the American biologist Rachel Carson cataloged the

environmental impact of spraying DDT and said it may cause cancer. It was eventually banned worldwide. New methods, such as intensive irrigation, were introduced to maximize productivity on poor farming land.

However, the new agricultural methods have generated further problems. Pesticides, especially earlier,

## America’s breadbasket

Resembling a work of modern art, green crops cover what was once shortgrass prairie in Kansas. The crops are fed from underground water reserves and by circular sprinklers, which cause the round growth pattern.

